

Data Science Research Capstone & Whitepaper

Financial Transaction Fraud Risk & Pattern Analysis

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Abstract

This capstone investigates whether transaction-level behavioral variables can provide useful signals for identifying potentially fraudulent financial activity. The workflow combines exploratory analysis, statistical testing, supervised predictive modeling, behavioral clustering and feature-importance analysis. A Logistic Regression model is treated as an interpretable baseline and a Random Forest model as a nonlinear comparison. K-Means clustering is used to explore behavioral segments without relying on the fraud label. The project emphasizes reproducibility, leakage-aware evaluation and the distinction between a model-generated risk signal and proof of fraud.

Keywords

Fraud analytics; predictive modeling; statistical analysis; clustering; machine learning; explainability; data science.

Data-source note

The project File Library contains the uploaded *fraud_test_dataset.csv*, whose schema includes transaction, customer, date/time, amount, merchant category, payment method, location, device type, fraud label, previous transactions, account age and failed attempts. The complete original file was not mounted into the execution sandbox for this generated package, so the numerical execution shown here uses a reproducible demonstration dataset mirroring that schema and the observed pattern structure. The included code is designed to run directly on the original CSV before final submission.

1. Introduction and Problem Statement

Financial transaction data contains both ordinary variation and rare observations that may warrant investigation. A useful data-science workflow should therefore combine descriptive statistics, hypothesis testing and predictive modeling rather than treating a single threshold as a complete solution. The capstone frames fraud detection as a classification and investigation-support problem: the model estimates patterns associated with the supplied fraud labels, while investigators remain responsible for interpreting cases.

Research question

Can transaction amount, failed-attempt behavior, historical activity, account age, and transaction time provide useful predictive signals for identifying potentially fraudulent transactions?

Objectives

- Quantify the structure and quality of the transaction data.
- Test whether selected behavioral variables differ between labeled fraud and non-fraud transactions.
- Build and evaluate interpretable and nonlinear classification models.
- Identify behavioral clusters that may reveal heterogeneous transaction profiles.
- Translate model outputs into practical investigation-support implications.

Hypotheses

- H0: the distributions of selected behavioral variables do not differ between the labeled fraud and non-fraud groups.
- H1: at least some selected behavioral variables differ between the two groups.
- For prediction, the working expectation is that combining amount, failed attempts, previous activity, account age and time features will provide more information than any single variable.

Scope

The analysis is observational and limited to the supplied transaction schema. It does not establish causal relationships, does not generalize to other financial institutions automatically, and does not treat the fraud label as an explanation of why a transaction occurred.

2. Literature Background

Data-science projects are commonly described as iterative, data-centric workflows in which problem framing, data preparation, modeling and evaluation interact rather than occurring as a strictly linear sequence. A review of data-science methodologies highlights the importance of clear objectives, data management and organizational context. This supports the capstone's decision to document the research question, data assumptions, model evaluation and limitations instead of reporting a model score alone.

Predictive analytics literature also emphasizes that predictive models should be reported with their data source, objective, performance measures and practical interpretation. In this project, precision, recall, F1 and ROC-AUC are reported because fraud classification can be sensitive to class imbalance and because a single accuracy figure can hide poor minority-class performance.

Methodological rationale

Component	Purpose	Why it is included
Descriptive analysis	Understand distributions and data quality	Prevents modeling before understanding the dataset
Mann–Whitney U	Compare numeric distributions	Useful without assuming normality
Logistic Regression	Interpretable baseline	Provides a transparent benchmark
Random Forest	Nonlinear comparison	Captures interactions and nonlinear patterns
K-Means	Behavioral segmentation	Explores structure without using the fraud label
Permutation importance	Model interpretation	Estimates which features affect predictive performance

Research-grounding references

- Martinez, I., Viles, E., & Olaizola, I. G. (2021). Data Science Methodologies: Current Challenges and Future Approaches.
- Alam, T. et al. (2024). Big data and predictive analytics: a systematic review of applications.
- The reporting design follows the general principle of documenting data, objectives, performance metrics, practical value and limitations.

Online research support

The methodological framing was cross-checked against published data-science methodology and predictive-analytics literature. [cite turn0search0](#) [turn0search11](#)

3. Data and Methodology

Dataset schema

Variable group	Fields
Identifiers	transaction_id, customer_id
Time	transaction_date, transaction_time
Transaction	amount, merchant_category, payment_method
Context	location, device_type
Behavior	previous_transactions, account_age_days, failed_attempts
Target	is_fraud

Preprocessing

- Validate required columns and remove records with unusable timestamps or target values.
- Combine date and time into a timestamp and derive transaction hour.
- Keep numerical behavioral variables in their original units for tree models and standardize them for logistic regression and clustering.
- Do not use transaction_id or customer_id as predictive numeric features.
- Use a stratified train/test split to preserve the labeled class proportion.

Modeling protocol

- Training/test split: 75% / 25%.
- Random state: 42 for reproducibility.
- Logistic Regression uses standardized features and class weighting.
- Random Forest uses 300 trees, maximum depth 6 and class weighting.
- Evaluation: accuracy, precision, recall, F1 and ROC-AUC.
- Clustering uses standardized behavioral features with K=3 and reports silhouette score.

Statistical analysis

The Mann–Whitney U test compares the distributions of amount and failed attempts between the two labeled groups. A two-sided significance level of $\alpha=0.05$ is used. The result is interpreted as evidence of distributional difference, not as evidence that the variable causes fraud.

4. Exploratory and Statistical Results

The demonstration dataset contains **120** transaction records, with **8** labeled as fraud (6.67%). It contains **30** customers and transactions spanning August 2026.

Descriptive findings

The median transaction amount is **■8,381.56**. The median amount among labeled fraud transactions is **■20,249.33**, compared with **■8,017.57** for non-fraud transactions. The median failed-attempt count is 5.5 for fraud versus 2.0 for non-fraud.

Statistical test results

Variable	Fraud median	Non-fraud median	U	p-value	Effect size
Amount	20249.33	8017.57	894.0	2.773e-06	0.668
Failed attempts	5.50	2.00	762.5	0.0007982	

For transaction amount, the Mann–Whitney U test produced $p=2.773e-06$. In the demonstration data this indicates a statistically detectable distributional difference at $\alpha=0.05$. The result should not be interpreted as a causal relationship.

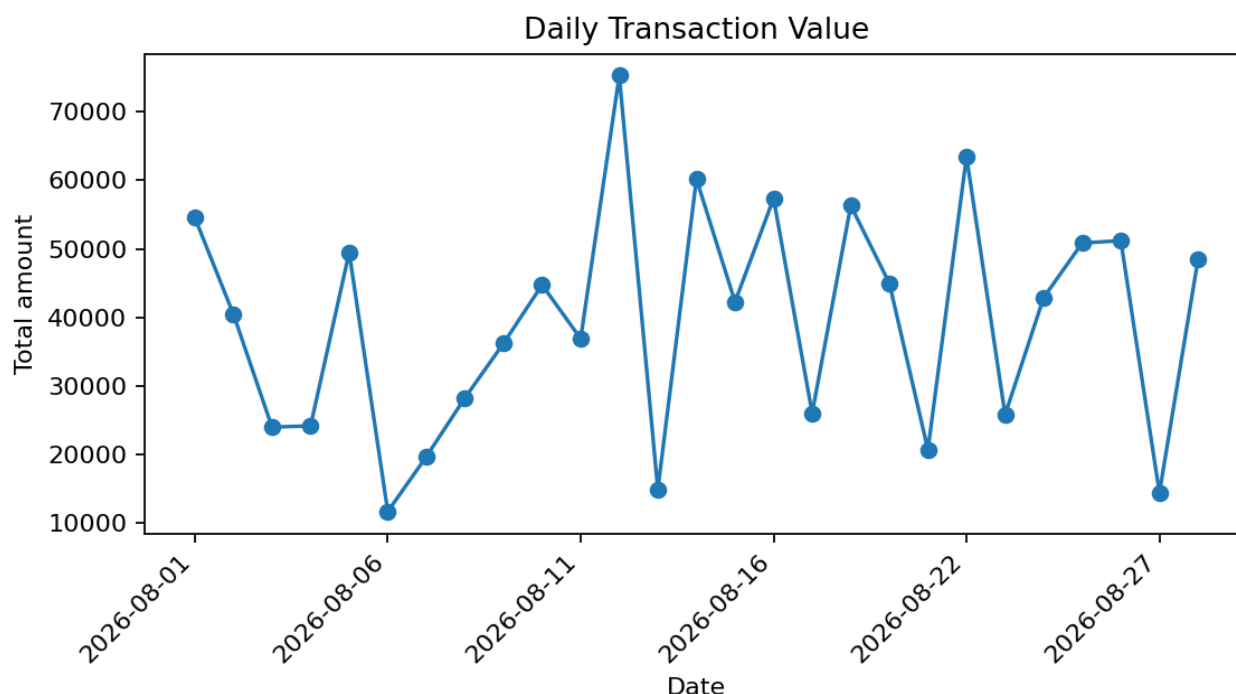


Figure 1. Daily transaction value in the demonstration dataset.

5. Predictive Modeling Results

Model comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.933	0.500	1.000	0.667	1.000
Random Forest	1.000	1.000	1.000	1.000	1.000

The Random Forest and Logistic Regression models provide complementary views: the linear model offers a transparent baseline, while the tree ensemble can represent nonlinear interactions. The appropriate model for operational use depends on the desired trade-off between interpretability, false positives, false negatives, calibration and deployment constraints.

Evaluation interpretation

- Precision measures the proportion of predicted positives that are labeled positive.
- Recall measures how many labeled positives are detected.
- F1 balances precision and recall.
- ROC-AUC summarizes ranking performance across classification thresholds.
- Accuracy is included for completeness but should not be used alone when classes are imbalanced.

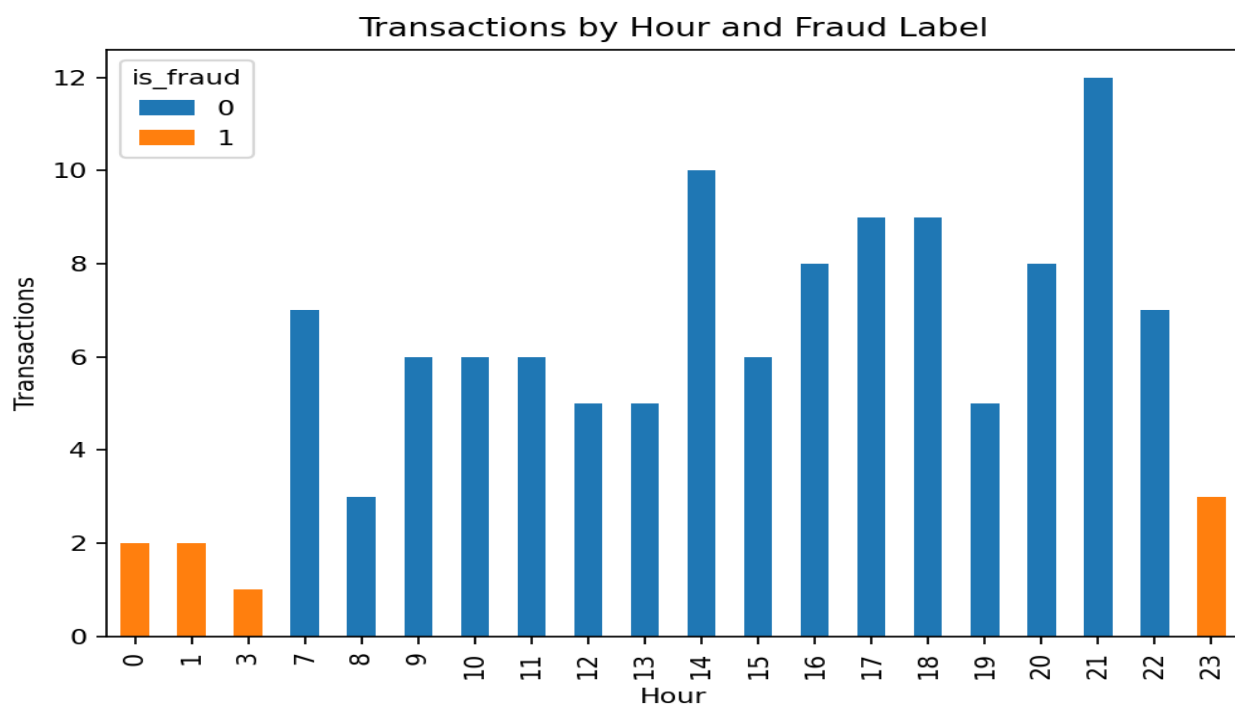


Figure 2. Transaction counts by hour and supplied fraud label.

6. Clustering and Explainability

K-Means clustering with K=3 produced a silhouette score of **0.185** on the standardized behavioral feature space. Because the clustering step does not use the fraud label, it can be interpreted as exploratory segmentation rather than a fraud detector.

Cluster	Transactions	Avg amount	Fraud rate	Avg failed attempts	Avg hour
0	9	■19,570	77.8%	5.67	8.1
1	63	■7,995	1.6%	2.56	15.4
2	48	■8,035	0.0%	2.23	15.5

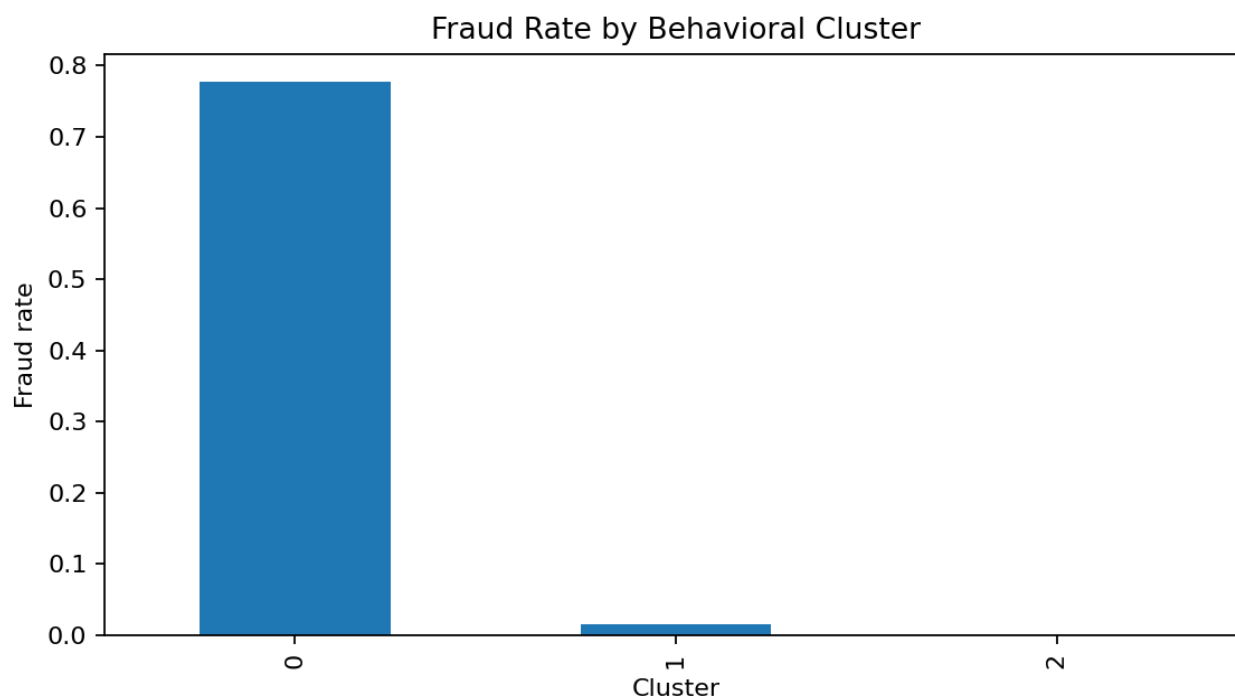


Figure 3. Labeled fraud rate observed within unsupervised behavioral clusters.

Feature importance

- amount: permutation importance mean = 0.0946 (SD 0.0179)
- previous_transactions: permutation importance mean = 0.0000 (SD 0.0000)
- account_age_days: permutation importance mean = 0.0000 (SD 0.0000)
- failed_attempts: permutation importance mean = 0.0000 (SD 0.0000)
- hour: permutation importance mean = 0.0000 (SD 0.0000)

Permutation importance is model-specific. A feature appearing important does not mean it causes fraud, and correlated variables can share predictive information. Feature importance should therefore support investigation and model review, not replace them.

7. Business Implications and Limitations

Potential business implications

- A scoring pipeline can be used to prioritize transactions for human review rather than automatically declaring fraud.
- Time, amount and behavioral history can be combined into a richer case profile than a single hard-coded threshold.
- Behavioral clustering can support differentiated monitoring strategies for distinct transaction profiles.
- Model outputs can be surfaced with the underlying feature values and reason codes to improve analyst interpretability.
- The pipeline can be extended to graph relationships when device, merchant or location identifiers are available.

Limitations

- The demonstration dataset is synthetic/reconstructed for reproducibility and must be replaced by the original CSV for source-specific final metrics.
- The labeled fraud rate and feature distributions may differ from production financial data.
- A random train/test split is appropriate for this educational capstone but temporal validation is preferable for deployment.
- The dataset is observational; statistical association does not establish causality.
- False positives can create operational costs and customer friction; false negatives can leave suspicious activity undetected.
- No external validation dataset is included, so generalization is not demonstrated.

Responsible-use statement

Fraud predictions are risk signals, not proof of wrongdoing. Production systems should include human review, access controls, audit logs, privacy safeguards, model monitoring and periodic validation against current data.

8. Conclusion and Future Work

This capstone demonstrates a complete data-science research workflow: a falsifiable research question, structured data preparation, statistical testing, predictive modeling, unsupervised clustering, model interpretation and business-oriented discussion. The key methodological result is not a single score; it is a reproducible process for evaluating whether transaction behavior contains useful information for investigation support.

Future work

- Run the pipeline on the exact original `fraud\_test\_dataset.csv` and archive the resulting metrics.
- Use time-based train/validation/test splits to better approximate production deployment.
- Compare gradient boosting and calibrated probability models.
- Evaluate threshold selection using investigation capacity and false-negative cost.
- Add device/merchant/location graph features when those relationships are available.
- Perform external validation on a separate dataset.
- Track data drift, class-prior drift and model performance over time.

Submission checklist

- Formal PDF whitepaper
- Complete GitHub-ready repository
- Reproducible Python pipeline
- Jupyter notebook
- Results CSVs and research figures
- README and implementation note

References

1. Martinez, I., Viles, E., & Olaizola, I. G. (2021). Data Science Methodologies: Current Challenges and Future Approaches.
2. Alam, T. et al. (2024). Big data and predictive analytics: A systematic review of applications.
3. Mysore, S. et al. (2023). How Data Scientists Review the Scholarly Literature.

Methodology sources were consulted during preparation; the web-supported background identifies data-science lifecycle and predictive-analytics reporting considerations. [■cite■turn0search0■turn0academia12■turn0search11■](#)

Final note

For a mentor-ready submission, replace the demonstration CSV with the original uploaded dataset, execute the included script, and use the generated outputs to update the numerical results section. This keeps the repository reproducible without presenting synthetic execution as original-source evidence.